SEIRS MultiSpecies MetaPopulation keywords / built-ins

```
% ---- SEIRSInfluenza.m
global mu gamma beta N nu sigma rho np ns;
np = 15;
ns = 2;
mu = repmat([0.0000365 0.00137], np, 1);
gamma = repmat([0.25 0.233], np, 1);
beta = repmat([0 0.025; 0 0.5], [1 1 np ]);
sigma = repmat([0.5 0.67], np, 1);
nu = repmat(0.00274, np, ns);
map = circshift(eye(np), 1); % one-way ring: flow from p to p+1
rho = cat(3,map*0.03,map*0.1);
N = repmat([5000 50000], np, 1);
I = zeros(np, ns);
p = 1;
s = 2;
I(p,s)=100;
S = N-I;
E = zeros(np, ns);
R = zeros(np, ns);

S = reshape(S,[1 ns * np ]);
E = reshape(E,[1 ns * np ]);
I = reshape(I,[1 ns * np ]);
R = reshape(R,[1 ns * np ]);
[T, Y] = resolve([S E I R]);

visualization(T,Y);
% ---- resolve.m
function [T, Y] = resolve(compartments)
tMax = 1500;
[T,Y]=ode45(@modelODEs,[0 tMax], compartments);
% ---- visualization.m
function res=visualization(T,Y)
global mu gamma beta N nu sigma rho np ns;
res=zeros(length(T),2);
res(:,1)=sum(Y(:,2*np*ns+1:2*np*ns+np),2);
res(:,2)=sum(Y(:,2*np*ns+np+1:3*np*ns),2);
plot(T,res');
legend('Human','Bird');
xlabel('Time');
ylabel('Total Infectious');
% ---- modelODEs.m
function res=modelODEs(t, compartments)
global mu gamma beta N nu sigma rho np ns ;
S = reshape(compartments(1: np*ns),[np ns]);
E = reshape(compartments(np*ns+1: 2*np*ns),[np ns]);
I = reshape(compartments(2*np*ns+1: 3*np*ns),[np ns]);
R = reshape(compartments(3*np*ns+1: end),[np ns]);
lambda = zeros(np, ns);
for p = 1:np
  for s = 1:ns
    lambda(p,s) = sum(beta(s,:,p).*I(p,:)./N(p,:));
  end
end
deltaS = zeros(np,ns);
deltaE = zeros(np,ns);
deltaI = zeros(np,ns);
deltaR = zeros(np,ns);
for s = 1:ns
  deltaS(:,s) = rho(:,:,s)*S(:,s)-sum(rho(:,:,s))'.* S(:,s);
  deltaE(:,s) = rho(:,:,s)*E(:,s)-sum(rho(:,:,s))'.* E(:,s);
  deltaI(:,s) = rho(:,:,s)*I(:,s)-sum(rho(:,:,s))'.* I(:,s);
  deltaR(:,s) = rho(:,:,s)*R(:,s)-sum(rho(:,:,s))'.* R(:,s);
end
dSdt = mu.*N + nu.*R - lambda.*S - mu.*S + deltaS ;
dEdt = lambda.*S - sigma.*E - mu.*E + deltaE ;
dIdt = sigma.*E - gamma.*I - mu.*I + deltaI ;
dRdt = gamma.*I - mu.*R - nu.*R + deltaR ;
dSdt = reshape(dSdt ,[1 np * ns ]);
dEdt = reshape(dEdt ,[1 np * ns ]);
dIdt = reshape(dIdt ,[1 np * ns ]);
dRdt = reshape(dRdt ,[1 np * ns ]);
res = [dSdt dEdt dIdt dRdt]';
```